OVERVIEW OF THE PROJECT

In today's digital-first world, an educational institution's reputation is no longer just built in the classroom—it is shaped online. While star ratings provide a quick snapshot of performance, they often fail to capture the " *why* " behind a student's experience. The project bridges that gap by deploying a modern data engineering pipeline that transforms raw Google Maps reviews into a strategic asset, allowing the institution to "listen" to its stakeholders at scale.

The solution moves beyond static analysis by establishing a living, breathing data ecosystem. We automated the data collection process using **N8N** and **Apify**, ensuring that every review is scraped and captured. To handle the complexity of human language, the project utilizes a high-performance computing environment on **Kaggle (T4 GPUs)** to run advanced Natural Language Processing tasks.

Understanding that data must be culturally relevant to be useful, the project moved beyond off-the-shelf solutions. I fine-tuned a **Llama 3** model specifically on a dataset of 42,000 names to accurately identify gender within an Indian context. Simultaneously, I leveraged the **CardiffNLP RoBERTa** model for precise sentiment scoring and **Google's Gemini 2.5 Flash-Lite** to act as an intelligent synthesizer **-** breaking down complex review text into tangible themes like "Infrastructure", "Academics" or "Campus Life."

The culmination of this engineering effort is a **Power BI** dashboard built on a robust Star Schema. The visualization layer does not just display data; it tells a story, empowering administration to move from reactive damage control to proactive institutional improvement.

OBJECTIVES

The core mission of the project is to bridge the gap between unstructured public feedback and strategic institutional decision-making. By moving away from manual, ad-hoc reviews, I aim to build a self-sustaining data ecosystem that transforms raw text into a clear competitive advantage. The following objectives outline my technical approach to achieving this robust, automated, and insight-driven system:

- **To Automate Data Acquisition:**

  Implement a reliable N8N workflow that interacts with the Apify Google Maps Scraper to fetch reviews automatically, eliminating manual data entry.

- **To Apply Advanced NLP for Sentiment Classification:**

  Utilize the RoBERTa architecture, specifically fine-tuned on social media text, to assign probabilistic sentiment scores (Positive, Neutral, Negative) to reviews, accounting for the nuances of informal language and emojis.

- **To Perform Demographic Inference:**

  Develop and deploy a fine-tuned Llama 3 classifier to predict the gender of reviewers based on Indian names, enabling diversity-focused sentiment analysis.

- **To Extract Actionable Themes:**

  Leverage the semantic reasoning capabilities of the Gemini 2.5 Flash Lite model to categorize reviews into meaningful topics and narrow them down to operational themes such as "Infrastructure," "Academics," and "Campus Life".

- **To Visualize Strategic Insights:**

  Construct a Power BI dashboard based on a Star Schema data model to present key performance indicators (KPIs) like Net Sentiment Score, Theme Distribution, and Temporal Trends to the college management.

SYSTEM OVERVIEW

The system is architected as a modular pipeline, where data flows sequentially through extraction, storage, processing, and visualization stages. This modularity ensures that individual components (like the sentiment model) can be upgraded without disrupting the entire workflow.

Scraping (Data Ingestion)

The system entry point is the N8N workflow. This module interacts with the Apify platform, specifically the "Google Maps Scraper" actor. Workflow is triggered on schedule or manually. It sends a request to Apify with the specific Google Maps URL or Place ID of Kongunadu Arts and Science College. The scraper navigates the dynamic web interface of Google Maps, handling pagination and scrolling to retrieve review objects including text, rating, timestamp, and reviewer metadata.

Storage (Data Lake)

Once scraped, the JSON data is parsed and routed to a centralized storage solution. Google Sheets serves as the immediate data lake, chosen for its accessibility and ease of integration. A dedicated Service Account, created in the Google Cloud Console, acts as the secure bridge. The service account is granted "Editor" permissions, allowing the N8N workflow to append new rows to the sheet systematically. This raw data includes noise, duplicate entries, and unstructured text, which are addressed in the subsequent phase.

Interpretation (Processing Core)

The heavy lifting occurs in the Kaggle Notebook environment. This module reads the raw data from Google Sheets and initiates a multi-step enrichment process:

- **Cleaning :** The indic_transliteration library transliterates and standardizes names (e.g., converting Tamil script names to English phonetics).
- **Gender Tagging :** The fine-tuned Llama 3 model processes the cleaned names. It classifies them into 'Male', 'Female', or 'Unknown'. The model was trained using Unsloth on a dataset of 42,000 labeled Indian names to handle cultural nuances effectively.
- **Sentiment Analysis :** The RoBERTa model scores each review. Logic is implemented to handle edge cases:

A. ***If a review has text***, the softmax scores from the model determine the sentiment.
B. ***If the text is empty***, the system falls back to the star rating (1-2 stars = Negative, 3 = Neutral, 4-5 = Positive).

- **Thematic Analysis :** The text is passed to the Gemini 2.5 Flash Lite API. A prompt engineering strategy instructs the model to extract specific "topics" (e.g., "Biotechnology Department") and aggregate them into broader "themes" (e.g., "Faculty & Teaching").

Gender Classifier (Fine-tuned Llama3 Model)

A custom gender classifier was developed by fine-tuning the Meta Llama-3 large language model on an Indian-name-specific dataset. The training was performed on Google Colab using a free-tier T4 GPU. The training framework employed the unsloth library for memory-efficient LoRA (Low-Rank Adaptation) fine-tuning, combined with the Hugging Face transformers and PyTorch libraries.

The training dataset (shisha-07/Indian_Names_with_Gender_Dataset, hosted on Hugging Face) was curated from Kaggle datasets and Gemini-assisted data generation, comprising 42,000 entries: 14,000 male names (label 1), 14,000 female names (label 2), and 14,000 neutral or non-name tokens (label 0). The final model was published to Hugging Face Hub (shisha-07/Llama-3-Indian-Gender-Classifier) in all standard formats including LoRA adapters, GGUF quantised weights, and a merged full-precision checkpoint. The source code and training scripts are available at the project GitHub repository.

Visualization (Business Intelligence)

The final processed data is exported to a CSV format (Maps_Reviews.csv) and loaded into Power BI. Here, a Star Schema is constructed to link the reviews (Fact Table) with dimensions like Time and Reviewer. This supports the creation of interactive reports that enable administrators to drill down from high-level sentiment trends to individual feedback entries.

SYSTEM ARCHITECTURE DIAGRAM